

Azure Databricks Retail Sales Pipeline Report

Project Overview

This project involved building an automated retail sales data pipeline using Azure Databricks, Azure Data Lake Storage, Delta Lake, and Unity Catalog. The objective was to ingest raw CSV sales data from Azure Data Lake Storage (ADLS), transform the data using PySpark, and store the processed data in Delta format for downstream analytics and reporting.

The solution was implemented using Databricks Workflows with serverless compute and Unity Catalog governance to simulate a modern enterprise-grade data engineering pipeline.

Data Pipeline Architecture

Architecture Overview

The pipeline follows a layered ETL architecture with orchestration through Databricks Workflows.

Azure Data Lake Storage (CSV Files)



Task 1: Ingestion Layer



Task 2: Transformation Layer



Task 3: Delta Storage Layer



Delta Lake Tables (Analytics Ready)

Pipeline Workflow

Task 1 – Data Ingestion

The ingestion notebook reads raw CSV sales data from ADLS using Spark DataFrames. The files are stored in an external location managed through Unity Catalog.

Task 2 – Data Transformation

The transformation layer performs business-level processing on the raw sales data.

Task 3 – Delta Lake Storage

The processed data is written into Delta Lake tables for analytics and reporting purposes.

Challenges Faced and Resolutions

1. ADLS Authentication Failure

Issue

While accessing CSV files from ADLS, the following error occurred:

Invalid configuration value detected for fs.azure.account.key

Root Cause

Databricks attempted to use legacy storage account key authentication instead of Unity Catalog governed authentication.

Resolution

- Removed legacy Spark configurations related to storage account keys
- Configured Unity Catalog storage credential correctly
- Verified managed identity permissions on ADLS
- Used external locations for governed access